

PYTHON • MySQL • POWER BI

Banking Analytics

An End-to-End Data Analyst Project

Raw transactional data se lekar ek six-page executive dashboard tak — Python me data cleaning, MySQL me relational analytics, aur Power BI me interactive visualization cover kiya gaya hai.

Data Cleaning

Database Design

SQL Analytics

Power BI Dashboards

Project Architecture

How data flows from raw files to business dashboards

1

Extract & Clean

Python (pandas)

- Read raw CSVs (customers, accounts, loans, transactions)
- Handle missing values & duplicates
- Parse dates, compute Age
- Export clean CSVs



2

Load & Model

MySQL (banking_analytics)

- SQLAlchemy + PyMySQL load
- Customers, Accounts, Loan, Credit_Cards, Transactions, Branches, Employees
- Joins, aggregates, segmentation queries



3

Visualize

Power BI

- Connects live to MySQL
- 6 interactive report pages
- Slicers, KPI cards, maps, trend charts

Step 1 — Data Cleaning with Python

pandas pipeline preparing raw banking data for MySQL

data_cleaning.py

```
customer.fillna(method='ffill', inplace=True)
account.fillna(0, inplace=True)
loan.fillna(0, inplace=True)
customer.drop_duplicates(inplace=True)

customer['DOB'] = pd.to_datetime(
    customer['DOB'], format='%d-%m-%Y', errors='coerce')

today = pd.Timestamp.today()
customer['Age'] = ((today - customer['DOB'])
                  .dt.days // 365)

customer.to_csv('customer_clean.csv', index=False)
```

1

Missing Values

Forward-fill customer records; zero-fill account & loan balances

2

Deduplication

Drop duplicate rows across customers, accounts, loans, transactions

3

Date Parsing

Convert DOB & Join_Date to datetime (day-first / explicit format)

4

Age Derivation

Compute customer Age from DOB using day-count arithmetic

5

Load to MySQL

SQLAlchemy + PyMySQL engine writes clean tables via to_sql()

Step 2 — MySQL Database: banking_analytics

Relational schema connecting customers to accounts, loans, cards & branches



Customers

Customer_ID, Name, Gender, Occupation, Annual_Income, DOB, Join_Date



Transactions

Transaction_ID, Account_ID, Date, Type, Amount, Payment_Method



Loan_Payments

Loan_ID, EMI_Amount, Payment_Status



Branchs

Branch_ID, Branch_Name, City, State



Accounts

Account_ID, Customer_ID, Branch_ID, Account_Type, Balance, Status



Loan

Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate, Status



Credit_Cards

Card_ID, Customer_ID, Card_Type, Credit_Limit, Available_Limit, Status



Employees

Employee_ID, Branch_ID, Salary

Step 2 — SQL Analytics Highlights

A sample of analytical queries written against banking_analytics

Customer Segmentation

CASE-based tiering (Premium / Gold / Silver / Standard) by summed account balance

Top 20 High-Value Customers

LEFT JOINS across Accounts, Loan & Credit_Cards, ranked by total balance

Branch-wise Performance

Customers, accounts & total deposits per branch via LEFT JOIN + GROUP BY

Inactive Accounts

NOT IN subquery flags accounts with no transaction in the last 6 months

Loan Default Risk

Loan_Status distribution with percentage share using a correlated subquery

Credit Utilization %

$(\text{Credit_Limit} - \text{Available_Limit}) / \text{Credit_Limit}$, ranked to flag high-risk cardholders

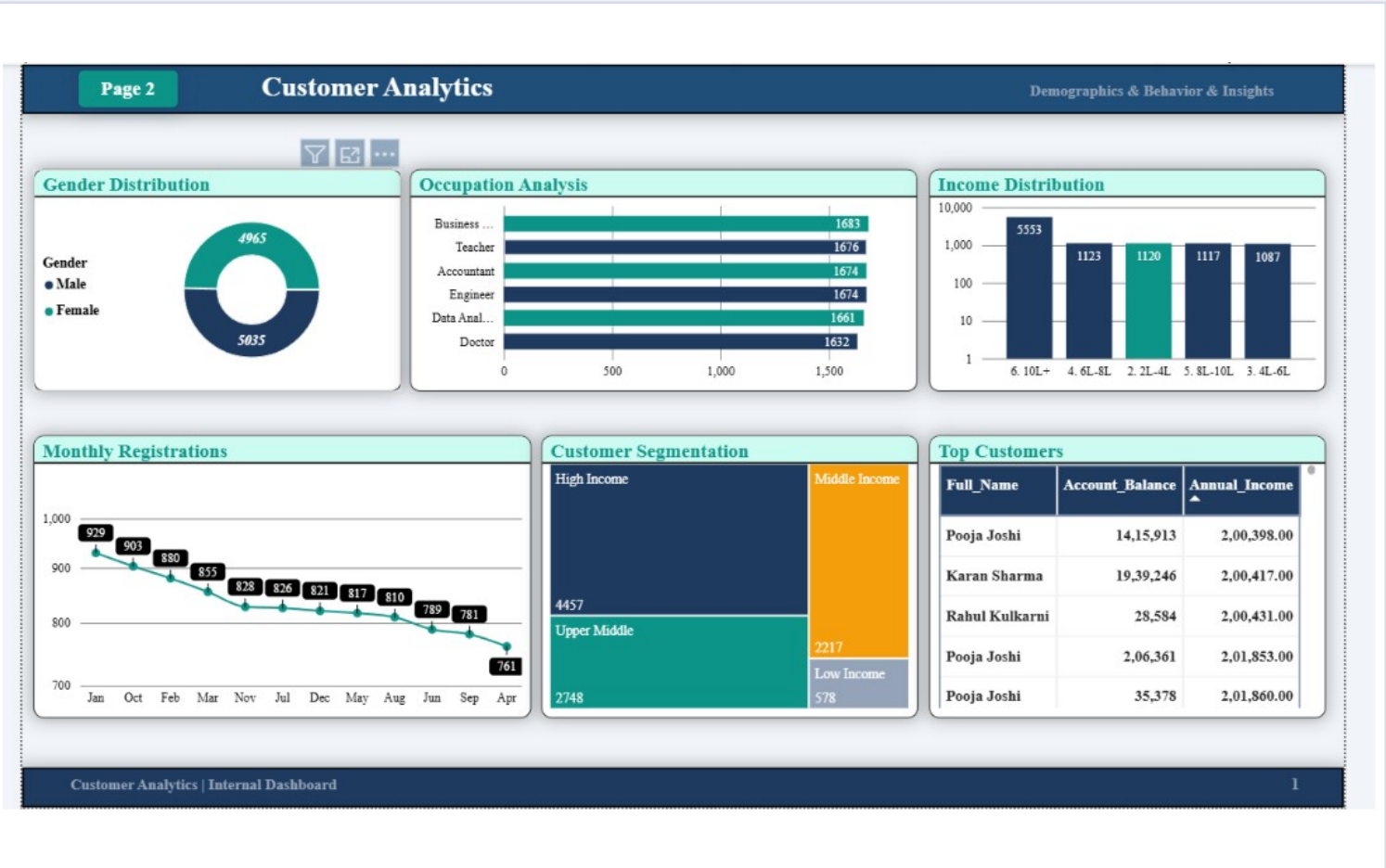
Power BI — Executive Summary Dashboard

High-level KPIs for management overview



Key Highlights

- 10K customers, 13K accounts, ₹7.25bn total balance
- 10K transactions, 10K loans, 10K credit cards tracked
- Monthly transaction trend & account-type mix (donut)
- Top 5 branches by deposits, with month slicer



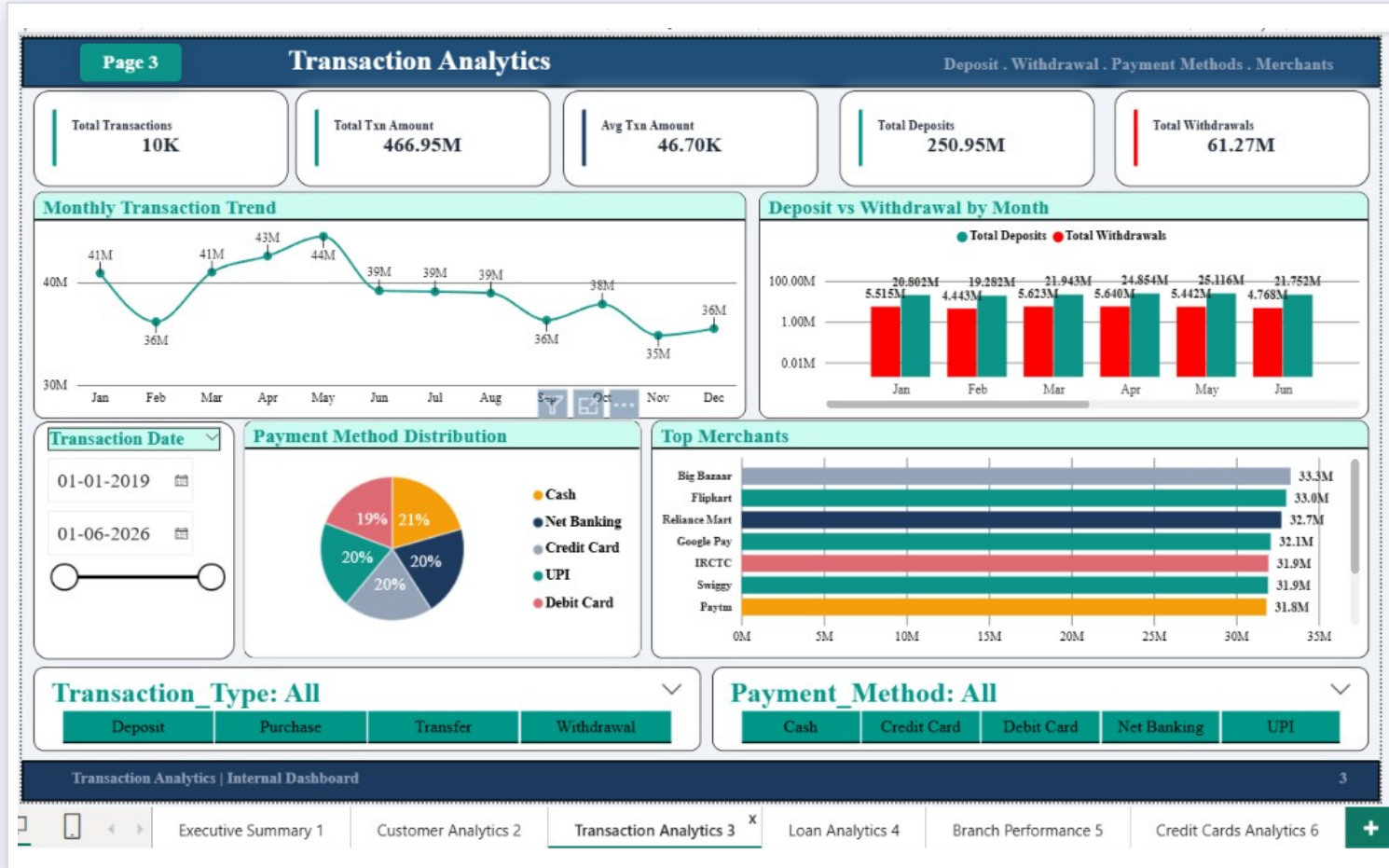
Key Highlights

- Gender split: 5,035 female vs 4,965 male customers
- Occupation analysis across Business, Teacher, Accountant & more
- Income distribution and customer segmentation treemap
- Top customers table ranked by account balance & income

Power BI — Transaction Analytics

Deposits, withdrawals, payment methods & merchants

Page 3



Key Highlights

- ₹466.95M total transaction value across 10K transactions
- Deposits (₹250.95M) vs withdrawals (₹61.27M) by month
- Payment method mix: Cash, Net Banking, Credit/Debit Card, UPI
- Top merchants by transaction value: Big Bazaar, Flipkart, Reliance Mart

Power BI — Loan Analytics

Loan type, status, EMI performance & default risk

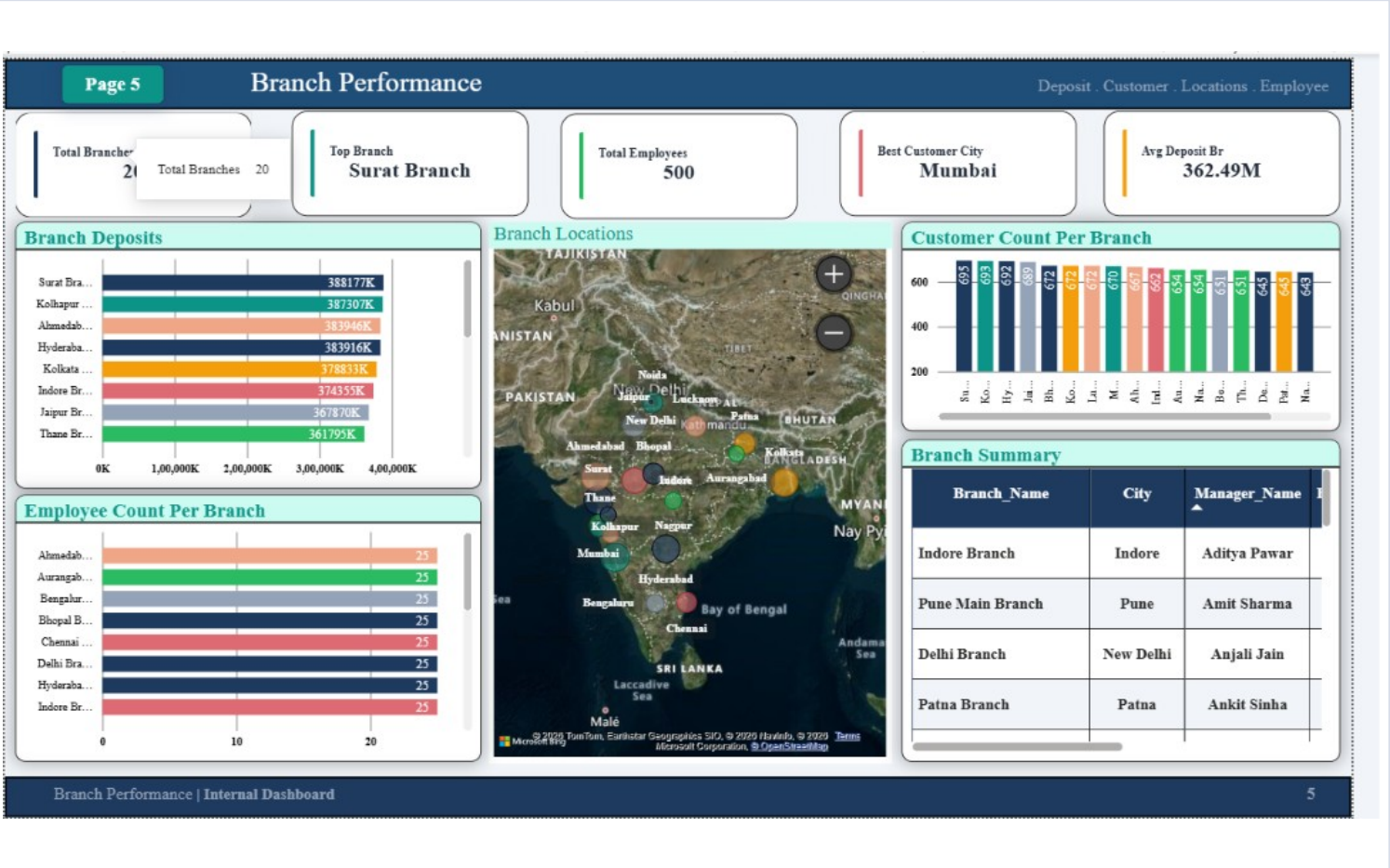


Key Highlights

- ₹23.61bn total loans issued, 4.86% default rate
- Loan mix by type: Home, Business, Car, Education, Personal
- Loan amount vs interest rate scatter by loan type
- EMI payment status (Paid / Missed / Pending) and top defaulters

Power BI — Branch Performance

Deposits, customers, employees & locations



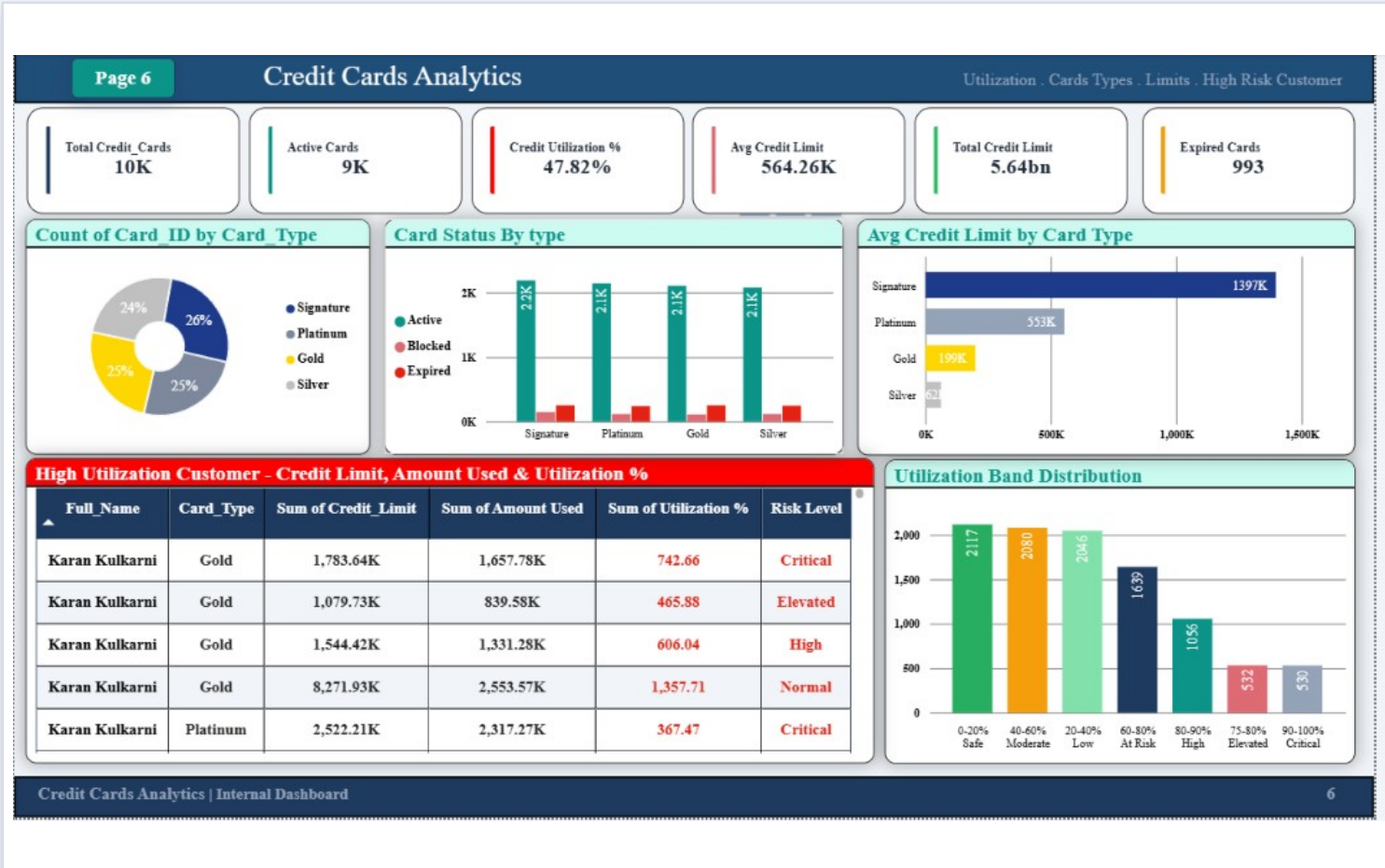
Key Highlights

- 20 branches, 500 employees, Surat Branch leads in deposits
- Mumbai is the best-performing customer city
- Map visual plots branch locations across India
- Branch summary table with manager names & city detail

Power BI — Credit Cards Analytics

Utilization, card types, limits & high-risk customers

Page 6



Key Highlights

- 10K credit cards, 47.82% average utilization, ₹5.64bn total limit
- Card type mix: Signature, Platinum, Gold, Silver
- Utilization band distribution flags Critical/Elevated risk tiers
- High-utilization customer table for risk monitoring

Key Takeaways

From messy CSVs to a decision-ready BI suite

Clean Pipeline

Python + pandas standardized dates, ages, duplicates & nulls before loading

Solid Data Model

8 normalized MySQL tables linking customers to accounts, loans, cards & branches

Actionable SQL

Segmentation, risk, utilization & performance queries ready for reporting

Live BI Reporting

6-page Power BI suite — Executive, Customer, Transaction, Loan, Branch, Credit Card

Thank You